Projected Quantum Kernels and the Geometric Test for Advantage

AI 631 Quantum Machine Learning, Fall 2026 — Project Synopsis (Project #6)

Farrukh Shahzad (25280034), Shams ul Arifeen (25280080), Haider Nadeem (25280003)
Lahore University of Management Sciences (LUMS)
Code: https://github.com/ifarrukhshahzad/AI631-Projected-Quantum-Kernels

***Abstract*—Huang *et al.* [1] proposed the geometric difference as a data-dependent indicator of when a quantum kernel can outperform a classical one, and introduced projected quantum kernels. We test these ideas at small scale: fidelity and projected quantum kernels versus tuned classical kernels on PCA-reduced Fashion-MNIST and on a relabeled dataset adapted from [1], in noise-free simulation, under a noise model, and in a small IBM hardware experiment.**

I. Introduction and Motivation

Quantum kernels measure data similarity using quantum states prepared by a data-encoding circuit, for use in a support vector machine (SVM) [2], [3]. A large feature space alone does not guarantee better learning: Huang *et al.* [1] showed that with enough training data classical models can often match quantum ones, and that the *geometric difference* between a classical kernel matrix K^C and a quantum kernel matrix K^Q helps assess the potential for a prediction advantage over classical kernels. They also introduced *projected quantum kernels* built from one-qubit reduced density matrices (1-RDMs).

II. Research Question

Question: At small scale (simulator and IBM hardware), is the geometric difference associated with when a fidelity or projected quantum kernel outperforms classical kernels? **Subquestions:** (1) how fidelity and projected kernels compare with tuned RBF, polynomial and linear kernels on original labels; (2) how g changes with qubit count and whether it tracks the accuracy gap; (3) whether the advantage indicated by a large g appears on the relabeled dataset; (4) how finite shots and noise change projected-kernel features and accuracy.

III. Falsifier

The geometric difference indicates only the *potential* for an advantage [1]; we test the project's empirical hypothesis under our setup, not the mathematical result of [1]. Each quantum kernel (fidelity, projected) is evaluated separately on the same splits as the classical models. For seed s, $\Delta A_s = A_s^Q - A_s^{C^*}$, where C^* is the classical kernel with the highest training cross-validation accuracy (the test set is used only for final evaluation); we report $\overline{\Delta A}$ and $\sigma_{\Delta A}$. As a project-specific heuristic (not a significance test), a quantum kernel shows an *improvement* if $\overline{\Delta A} > \sigma_{\Delta A}$; failing this means no improvement was demonstrated, not that equality or inferiority is proved. g_λ (Sec. IV-B) is treated as a continuous variable from training data; any thresholds used for the conditional falsifier will be specified using training-data diagnostics before test accuracies are examined. The hypothesis is unsupported if (i) on the engineered dataset, where $g_\lambda(K^{C^*} \| K^Q)$ is large relative to the original task, no quantum kernel shows an improvement; or (ii) on the original labels, where g_λ is small, a quantum kernel shows an improvement over C^*. If no setting with a clearly larger g_λ is obtained, (i) is reported as untested (inconclusive).

IV. Methods

A. Kernels and Encoding

Encoding (proposed): features are scaled to $[-1, 1]$ (fitted on training data); a block of $R_y(\alpha x_i)$ on each qubit i followed by a CNOT chain between neighbours is repeated L times from $|0\rangle^{\otimes n}$. The main setting is $L = 2$; $L = 1$ is a baseline, because for one layer the data-independent CNOT block cancels in the overlap and the fidelity kernel reduces to $\prod_i \cos^2\big(\alpha(x_i - x_i')/2\big)$, which is classically easy (this cancellation does not apply to the projected kernel). $L = 2$ is not claimed to give any computational advantage. Both quantum kernels share the circuit architecture; α is tuned per kernel by training cross-validation. Qubits and PCA features are varied jointly (n = 2–8) with L and connectivity fixed; this sweep also changes gate counts, which we report.

Fidelity kernel: ideally $k^Q(x, x') = |\langle\psi(x)|\psi(x')\rangle|^2$. With finite shots or noise we use the estimator $\hat{k}^Q$ = probability of the all-zero outcome of the compute–uncompute circuit $U(x')^\dagger U(x)$; under noise this is a circuit probability, not the fidelity of two mixed states.

Projected kernel: the 1-RDM of qubit k is

$$\rho_k(x) = \tfrac{1}{2}\big(I + \langle X_k\rangle_x X + \langle Y_k\rangle_x Y + \langle Z_k\rangle_x Z\big), \quad (1)$$

$$k^{PQ}(x, x') = \exp\Big(-\gamma \sum_k \big\|\rho_k(x) - \rho_k(x')\big\|_F^2\Big), \quad (2)$$

where $\|\rho_k(x) - \rho_k(x')\|_F^2 = \frac{1}{2}\sum_{P\in\{X,Y,Z\}}\big(\langle P_k\rangle_x - \langle P_k\rangle_{x'}\big)^2$. For a general one-qubit state, X, Y and Z determine the 1-RDM. For our ideal R_y–CNOT encoding, amplitudes are real and Y expectations vanish; we retain Y measurements as a consistency check under sampling and device noise. Measurement: Z directly; X after H; Y after applying $S^\dagger$, followed by H. Each basis circuit gives that Pauli for all

qubits, so features are computed once per sample and reused for every pair.

Classical baselines: RBF, polynomial and linear kernels, tuned by training cross-validation.

B. Geometric Difference

The ideal definition [1], with kernels normalised to trace N, is $g = \sqrt{\|\sqrt{K^Q}(K^C)^{-1}\sqrt{K^Q}\|_\infty}$. Since K^C can be singular (the linear kernel is rank-deficient when samples far exceed features), we compute the regularised diagnostic

$$g_\lambda(K^C\|K^Q) = \sqrt{\left\|(K^Q)^{1/2}\,(K^C+\lambda I)^{-1}(K^Q)^{1/2}\right\|_2}, \quad (3)$$

with principal square roots by eigendecomposition. λ is fixed by a predefined numerical rule, never by test accuracy: the smallest value in $\{10^{-6},\dots,10^{-2}\}$ for which the condition number of $K^C+\lambda I$ is below 10^8; the same rule is used in label generation, and sensitivity of g_λ to λ is reported. g_λ is computed on training samples, from ideal (noise-free) quantum kernels only. The primary analysis plots each quantum kernel's ΔA against $g_\lambda(K^{C^*}\|K^Q)$ per seed; all six pairs {fidelity, projected}$\times${RBF, polynomial, linear} are reported separately.

C. Engineered Dataset (adapted from [1])

We adapt the construction in the supplementary information of [1]. On the full subsample, with fixed hyperparameters, K^C is a fixed RBF reference and K^Q the projected kernel; we take the top eigenvector v of $(K^Q)^{1/2}(K^C+\lambda I)^{-1}(K^Q)^{1/2}$, form $y=(K^Q)^{1/2}v$, and threshold at the median (ties broken deterministically by sample index) to get balanced binary labels. Labels are generated separately for each n, so each qubit count defines its own task. PCA and scaling for label generation are fitted once on the full subsample, only to define the task; labels are frozen before the split, and evaluation refits PCA and scaling on training folds only. This is a controlled, dataset-dependent benchmark, not an independently labelled task; the RBF reference, median thresholding and refitting are simplifications, so any advantage must be shown empirically.

D. Data, Noise and Protocol

Fashion-MNIST [4], T-shirt/top vs. Shirt (visually similar). Proposed size: 300 training (150 per class) and 100 test (50 per class) images; 5 seeds, each varying the subsample, the stratified split (by original or engineered labels) and shot sampling; 5-fold cross-validation inside training data. Experiments use Qiskit and the course NB11 capstone harness for folds, leakage-free preprocessing, baselines, ablations and noise sweeps. The proposed noise model is a backend-derived Qiskit Aer model (no calibration values are assumed here). Finite-shot ideal and noisy simulations use matched shot budgets, the same samples and encoding parameters; noisy results include both sampling error and modelled device errors. Estimated fidelity matrices are checked for symmetry and positive semidefiniteness, and any repair (eigenvalue clipping) is disclosed; projected kernels built from one cached feature vector per sample need no repair.

Reported: test accuracy (mean $\pm$ std over seeds); paired ΔA vs. g_λ as a function of n; ideal vs. finite-shot vs. noisy kernels; qubits, transpiled depth, two-qubit gates, fidelity cost per evaluated pair, projected-feature cost per sample, and total circuits and shots.

E. IBM Hardware Component (proposed)

For about 10 test samples, $\langle X_k\rangle$, $\langle Y_k\rangle$, $\langle Z_k\rangle$ are estimated for all qubits (3 basis circuits per sample, about 4,000 shots each, roughly 120,000 shots per configuration excluding repeats). These are compared with exact, finite-shot and noisy simulator values for the same samples, validating projected-kernel features (not hardware classification). Backend, job IDs, layout, transpiled depth and any mitigation are reported. **Risks:** simulation cost (at most 8 qubits, smaller samples if needed), ill-conditioned K^C (rule above), hardware queues (submitted early in week 3).

V. Work Split and Plan

Ownership: Farrukh — fidelity kernel and IBM hardware run; Shams — projected (1-RDM) kernel and noise model; Haider — classical kernels, geometric difference and engineered dataset. All first agree on a shared interface (data, scaling, encoding, normalisation, seeds, split IDs).

- **Week 1 (Nov 9–15):** Haider: data loading, PCA, fixed folds and seeds, classical SVM baselines. Farrukh: encoding circuit, fidelity kernel on 2–4 qubits, IBM test job. Shams: 1-RDM extraction from simulated states. Report: introduction and methods.
- **Week 2 (Nov 16–22):** Shams: projected kernel and bandwidth tuning; noise-model kernels. Haider: geometric difference and the relabeled dataset. Farrukh: qubit-count and shot-count studies for the fidelity kernel. Report: results draft with accuracy-vs-g figures.
- **Week 3 (Nov 23–29):** Farrukh: hardware estimation of Pauli X, Y, Z values for about 10 samples. Shams: simulator values for the same samples and the four-level comparison. Haider: final g-vs-accuracy analysis. All: report draft and talk.
- **Week 4 (Nov 30–Dec 6):** final report and tagged GitHub release by December 6.

Presentation: on the assigned date, November 29 or December 6; each member presents their part. **Deliverables:** IEEE LaTeX report (PDF and source); GitHub repository with a README reproducing every figure and table, requirements, fixed seeds, saved kernels and results, and commits from every member.

References

[1] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, and J. R. McClean, "Power of data in quantum machine learning," *Nat. Commun.*, vol. 12, p. 2631, 2021.

[2] M. Schuld and N. Killoran, "Quantum machine learning in feature Hilbert spaces," *Phys. Rev. Lett.*, vol. 122, p. 040504, 2019.

[3] V. Havlíček *et al.*, "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, pp. 209–212, 2019.

[4] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms," arXiv:1708.07747, 2017.